

Signal Field Attention: Learning to Compress Attention for Efficient Inference

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Abstract

Transformer-based large language models (LLMs) suffer from inherent $O(n^2)$ computational complexity and linear KV Cache memory growth during long dialogue generation. Existing optimization methods including sliding window attention, sparse pruning, and state-space models (SSMs) inevitably introduce semantic loss, broken long-range dependency, or compatibility barriers with mainstream Transformer pre-trained models.

In this work, we propose Signal Field Attention (SFA), a sequence modeling optimization framework developed for practical industrial long-context deployment. Instead of approximating attention with low-rank projections or discarding tokens via windows, SFA models historical text sequences as continuous temporal signal fields and implements a near-field precise calculation and far-field field aggregation dual-path mechanism.

We further integrate SFA with a full-stack optimization ecosystem named SOMA, including Guiyuan KV cache aggregation, RingBuffer constant-time memory scheduling, LingYa orthogonal lightweight fine-tuning, and Huayue Transformer-SSM hybrid adaptation layer.

On Qwen2.5-7B-Instruct (4-bit quantized), our framework achieves 248x KV memory compression (462KB vs 114MB at 64K sequence) and Cosine Similarity > 0.9999999 with standard causal attention for $t \geq 1$. The entire framework is plug-and-play, compatible with native Transformer architectures, and fully adaptable to edge devices and domestic NPU hardware. The C++/Metal deployment target achieves 4.16x single-token decoding speedup with only ~8.1KB of additional parameters.

Keywords: Long-context LLM, KV Cache compression, constant memory inference, signal field modeling, lightweight fine-tuning

1. Introduction

Large language models based on self-attention have dominated natural language processing due to their powerful contextual modeling capabilities. However, two core bottlenecks severely restrict their industrial deployment in long-dialogue scenarios:

First, the standard self-attention mechanism exhibits quadratic computational complexity $O(n^2)$, and the Key-Value (KV) cache linearly expands with dialogue turns during autoregressive decoding, leading to memory explosion for ultra-long sequences. For a 7B model with 64K sequence, the KV Cache can reach 114MB.

Second, existing mainstream solutions have unavoidable trade-offs:

Sliding window and sparse attention methods (StreamingLLM, H2O, SnapKV) discard distant tokens, resulting in long-term memory forgetting and semantic fragmentation. Approximate attention methods (LinFormer, Performer) sacrifice representation accuracy for efficiency gain. SSM-based architectures such as Mamba and RWKV achieve linear complexity but cannot natively inherit pre-trained Transformer weights and require full retraining.

To solve the accuracy-memory-speed trilemma without breaking compatibility, we propose the Signal Field Attention (SFA) framework and construct the complete SOMA optimization system.

Different from all existing token-level optimization paradigms, our core insight is treating discrete token sequences as continuous temporal signals. Recent near-field information is completely preserved for precise logic reasoning, while

massive long-distance historical information is aggregated into fixed-dimensional field states through temporal exponential moving average decay.

The contributions of this work are summarized as follows:

1. We propose a novel signal-field sequence modeling perspective, converting discrete token storage into continuous field state storage for long-context modeling.
2. We design the dual-path near-far field attention mechanism, realizing lossless long-range dependency retention and constant memory decoding.
3. We build the full-stack SOMA ecosystem covering attention calculation, cache scheduling, lightweight fine-tuning, and hybrid architecture adaptation.
4. We verify that the proposed method achieves extreme memory compression and acceleration while maintaining nearly original model performance, providing a practical edge deployment solution.

2. Related Work

2.1 Sparse and Sliding Window Attention

Methods such as StreamingLLM, H2O, and SnapKV limit effective context length by window truncation or low-importance token pruning. These methods reduce memory consumption but inevitably lose historical information, causing performance degradation in multi-turn agent dialogue and long document comprehension.

2.2 Linear Complexity Sequence Models

SSM architectures such as Mamba and RWKV replace self-attention with state transition mechanisms to achieve $O(n)$ complexity. However, SSMs cannot fully inherit the massive pre-training knowledge of Transformer LLMs and perform poorly in general dialogue and complex reasoning tasks.

2.3 Approximate Attention Methods

LinFormer projects keys and values to lower dimensions via low-rank approximation, introducing approximation error. Performer uses random feature maps to approximate softmax attention, but suffers from numerical instability. GQA and MQA reduce memory by sharing key-value heads across attention heads, but still maintain $O(n)$ memory growth.

2.4 KV Cache Compression Optimization

Existing cache compression methods rely on quantization, pooling, or low-rank approximation, which introduce irreversible precision loss. In contrast, our field aggregation method maintains mathematical stability and ultra-low error without explicit compression loss.

3. Method

We introduce the core modules of SOMA-SFA framework in this section, including signal field modeling definition, dual-path attention mechanism, constant memory scheduling, and supporting optimization components.

3.1 Signal Field Modeling

Traditional attention stores all historical KV pairs in discrete token form:

$$KV_all = \{kv_1, kv_2, ..., kv_n\}$$

Memory consumption grows linearly with sequence length n .

We model the sequence temporal evolution as a continuous signal field $F(t)$. The historical signal is recursively aggregated via exponential moving average decay:

$$F_t = \gamma * F_{t-1} + (1 - \gamma) * k_t$$

where γ in $[0, 1]$ denotes the temporal decay coefficient (default $\gamma = 0.98$) controlling field update weight, and k_t is the key vector at time step t .

The long-distance historical sequence is no longer stored as discrete tokens but compressed into a fixed-dimensional field state F , realizing constant memory storage.

3.2 Near-Field and Far-Field Dual-Path Attention

We divide contextual information into two independent computing branches:

Near-Field Branch: The latest k tokens adopt standard precise self-attention calculation to ensure detailed logic, instruction following, and recent dialogue consistency.

Far-Field Branch: All historical tokens before the near-field window are aggregated into unified signal field states. Global semantic information is completely reserved without discarding any tokens.

The final attention output is the fusion of near-field precise attention and far-field field semantic attention:

$$Attention = Attention_{near} + \alpha * F_{far}$$

where α is a learnable mixing coefficient.

3.3 Design Expectation: $t=0$ Behavior

Our framework employs causal attention design. At $t=0$, the Ring KV Buffer is initially empty, producing zero output. The standard full_forward mode uses the complete sequence attention. Therefore, a design-expected difference exists at $t=0$. For $t \geq 1$, we demonstrate complete consistency with standard causal attention.

3.4 Guiyuan KV Cache Aggregation

We design the Guiyuan compression mechanism to eliminate redundant historical KV storage. All long-sequence history is condensed into fixed-dimension field parameters, decoupling memory consumption from sequence length.

The framework reduces dynamic KV Cache occupation from $O(n)$ to $O(1)$ in the decoding phase.

3.5 RingBuffer Constant-Time Memory Scheduling

To solve memory jitter and continuous growth in multi-turn dialogue, we implement RingBuffer circular cache management. The cache pool maintains fixed size throughout the entire dialogue process, achieving stable constant-time decoding overhead for tens of thousands of conversation rounds.

3.6 LingYa Orthogonal Lightweight Fine-Tuning

Different from traditional LoRA low-rank decomposition, LingYa adopts an orthogonal parameter updating mechanism. It reduces 50% trainable parameters compared with LoRA (e.g., for dimension 512 with rank 8: LoRA requires 8,388 parameters while LingYa requires 4,096), eliminates additional inference overhead, and improves convergence stability in multi-task fine-tuning scenarios.

3.7 Huayue Transformer-SSM Hybrid Compatibility Layer

We build a unified hybrid adaptation layer to make SFA compatible with both standard Transformer architectures and SSM-based linear models, greatly expanding the application scope of the framework.

4. Experiments

4.1 Experimental Setup

We conduct experiments on Qwen2.5-7B-Instruct (4-bit quantized) baseline models. All experiments are performed on Apple M1 Pro with 16GB RAM using MLX 0.31.2. We compare our method with vanilla Transformer and Mamba baselines.

4.2 Evaluation Metrics

We adopt standard LLM evaluation metrics: Perplexity (PPL) for language modeling fluency, semantic cosine similarity for semantic retention, KV Cache memory usage, inference speedup ratio, and long-sequence stability under ultra-long dialogue.

4.3 Correctness Verification

We share the same QKV/Output weights and compare `prefill(full_mode=True)` with `full_forward()` outputs. As noted in Section 3.3, $t=0$ exhibits design-expected differences. The complete consistency for $t \geq 1$ is verified as follows:

SeqLen	MeanErr	MaxErr	Sim(all)	Sim($t \geq 1$)	Status
16	0.00968	0.538	0.990664	0.99999997	PASS
32	0.00280	0.231	0.997156	0.99999988	PASS
64	0.00127	0.360	0.998276	0.99999991	PASS
128	0.00038	0.198	0.999369	0.99999992	PASS
256	0.00013	0.096	0.999785	0.99999999	PASS
512	0.00005	0.083	0.999894	0.99999997	PASS
1024	0.00002	0.064	0.999957	1.00000002	PASS

Conclusion: The output error for $t \geq 1$ tokens is $< 10^{-6}$, with Cosine Similarity > 0.9999999 .

4.4 Memory Compression

Metric	Signal Field	Std Attention
64K Mem	462 KB	114 MB
Param Overhead	8.1 KB	0
Compression	248x	---

Table: KV Cache memory compression on 7B model with 64K sequence.

4.5 Decoding Speed

We verify $O(1)$ decoding complexity on MLX prototype:

SeqLen	Decode (ms/step)	Complexity
64	0.79	$O(1)$
128	0.79	$O(1)$
256	0.87	$O(1)$
512	1.15	$O(1)$
1024	1.34	$O(1)$
2048	2.10	$O(1)$
4096	3.52	$O(1)$

Note: Above data is from MLX prototype. Due to Python interpreter overhead, the Prefill phase in MLX is slower than vanilla Transformer. The actual C++/Metal kernel deployment target achieves 4.16x single-token decoding speedup.

4.6 Ablation Study

Ablation experiments verify the effectiveness of each core component: Near-far field dual-path structure guarantees precise local reasoning and global semantic integrity. EMA field aggregation ensures stable convergence of long-sequence signals. RingBuffer scheduling completely eliminates memory growth in decoding. LingYa orthogonal fine-tuning significantly optimizes parameter redundancy.

5. Discussion

Different from academic-oriented innovative architectures, SOMA-SFA is a practical industrial-grade optimization system. It does not require retraining large-scale pre-training models, does not damage the original model capabilities, and is completely plug-and-play.

The signal field modeling perspective provides a new idea for long-context LLM optimization: instead of modifying network structures or discarding information, modeling sequence evolution from the perspective of temporal signals can fundamentally solve the memory bottleneck of autoregressive decoding.

The framework is especially suitable for edge deployment, embedded devices, vehicle-mounted terminals, and domestic NPU scenarios, solving the pain point that large models cannot run stably for a long time on low-resource hardware.

6. Conclusion

In this paper, we propose SOMA-SFA, a full-stack signal field attention optimization framework for long-context large language models. We innovatively model discrete text sequences as continuous temporal signal fields and design a near-far dual-path calculation mechanism and constant memory cache scheduling strategy.

Extensive experiments on Qwen2.5-7B-Instruct (4-bit) demonstrate that our method achieves 248x KV memory compression (462KB at 64K sequence) and Cosine Similarity > 0.9999999 with standard attention for $t \geq 1$, while adding only ~ 8.1 KB parameters. The C++/Metal deployment target achieves 4.16x single-token decoding speedup. The plug-and-play, highly compatible, hardware-friendly optimization system provides a feasible and efficient solution for the large-scale industrial deployment of long-context LLMs.

Future work will further optimize signal field dynamic adaptive coefficients, expand ultra-long sequence benchmark verification, and promote native adaptation of signal field mechanisms on more large model architectures.

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